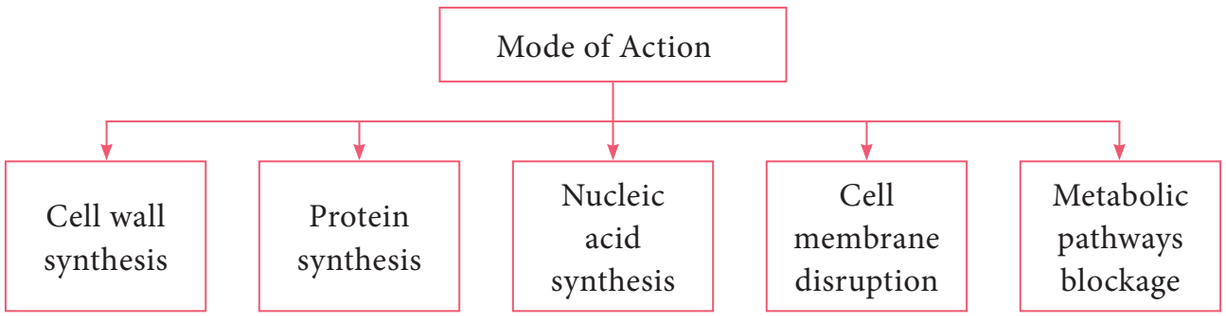


Flowchart 3.1: Mode of action of antibiotics

cell walls have a unique structure which is not found in eukaryotic cells. The important cell wall attacking drugs are Penicillin, Cephalosporin, Ampicillin, Methicillin and Vancomycin.

2. Inhibition of Protein Synthesis

Many therapeutic antibiotics discriminate between prokaryotic and eukaryotic ribosomes and inhibit protein synthesis. The therapeutic index of these drugs is fairly high, but not as favourable as that of cell wall synthesis inhibitors. Several of these drugs are medically useful and effective research tools because they block individual steps in protein synthesis. Some therapeutic drugs bind to 30S while others attach to 50S ribosomal subunits. Example Streptomycin, Chloramphenicol, Tetracyclin and Erythromycin

3. Inhibition of Nucleic Acid Synthesis

Some antimicrobial drugs or antibiotics inhibit nucleic acid synthesis. These are not selectively toxic as other drugs. This is due to the fact that prokaryotic and eukaryotic nucleic acid synthesis mechanisms do not vary greatly. Example Quinolones, Novobiocin, Actinomycin and Rifampin



A chemotherapeutic agent destroys or inhibit the intracellular parasite by penetrating the cells and tissues of the host in effective concentrations.

4. Disruption of Cell Membrane

There are some antimicrobial drugs or antibiotics that act as cell membrane disorganizing agents. Polymixins are such drugs of clinical importance.

E.g. Polymixin B and Polymixin E (colistin)

5. Blocking Metabolic Pathways

Some therapeutic drugs act as antimetabolites and block the functioning of metabolic pathways. They competitively inhibit the key enzymes in the metabolic pathway. Example Sulfonamides, Trimethoprim, Dapsone and Isoniazid (Figure 3.2).

3.7 Antimicrobial Susceptibility Testing

Antimicrobial susceptibility tests are used to determine the type and quantity of antimicrobial agents used in chemotherapy. One of the most important functions of a clinical laboratory is to determine the antimicrobial susceptibility

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